

Microsoft AI Upskilling

Build, evaluate, and govern enterprise-grade AI agents on Microsoft Foundry

A hands-on, three-session program • runnable code, not just slides

A fully Microsoft-led, architect-driven training program on Microsoft Foundry - six intensive sessions that take participants from agentic architecture principles to real, running code. Every module is delivered by senior Microsoft architects through live demos and reproducible exercises, avoiding generic theory. The entire journey is structured around a unified business scenario: **AdvertSphere Broadcasting**, a fictional advertising concessionaire across TV, radio, streaming, digital, and social.

What you'll take home

A solid mental model of the Microsoft agent platform, plus twelve hands-on labs and reference code ready to adapt to your own projects.

- Build an agent on Microsoft Foundry, extend it with an MCP tool, and call it from code through the Responses API.
- Evaluate it rigorously: groundedness, content-safety, and custom evaluators, run both locally and as cloud jobs.
- Pressure-test it before your users do: synthetic and adversarial data generation, jailbreak and red-teaming simulations.
- Compose and connect agents with the Microsoft Agent Framework and open protocols (MCP, A2A), with identity and governance through Entra Agent ID and Agent 365.

The three sessions

Each session: 8 hours over two consecutive days — Day 1 vision & architecture (leadership + technical), Day 2 hands-on labs (technical roles).

1 • Microsoft AI Platform

Foundry architecture; build, ground, and govern agents; Agent 365. Day-2 labs: prompt agent → MCP tool → Responses API → publish to Agent 365.

2 • GenAI Evaluation

Metrics, local and cloud evaluation, custom evaluators, red teaming. Day-2 labs: local eval → dataset generation → cloud eval → red teaming.

3 • Agentic Frameworks

From Semantic Kernel to Agent Framework, MCP, A2A, and agent-fleet governance. Day-2 labs: build and connect agents across open protocols.

3 sessions • 6 four-hour half-days • 12 hands-on labs • 1 real end-to-end scenario

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For learning purposes only — personal and unofficial. Authoritative, maintained reference: github.com/microsoft/foundry/foundry-samples